

Computer Vision Final Project

Stereo Matching

National Taiwan University
Spring 2026

Introduction of Stereo Matching

- Q: How do human's eyes judge the **distance between two objects** or the **depth of object**?
- A: Two eyes perceive same object slightly differently and our brain can merge two images into a 3D image.
- That's Stereo Matching!!

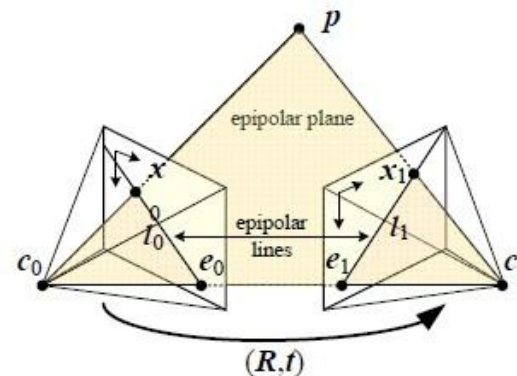
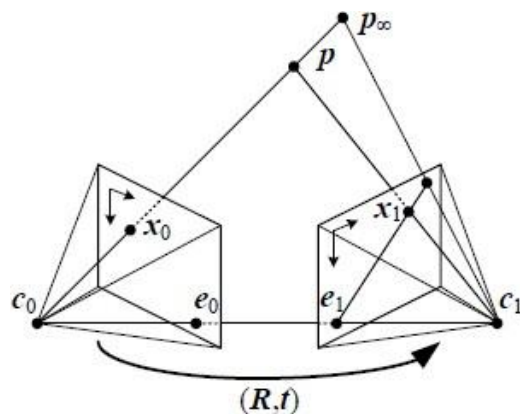
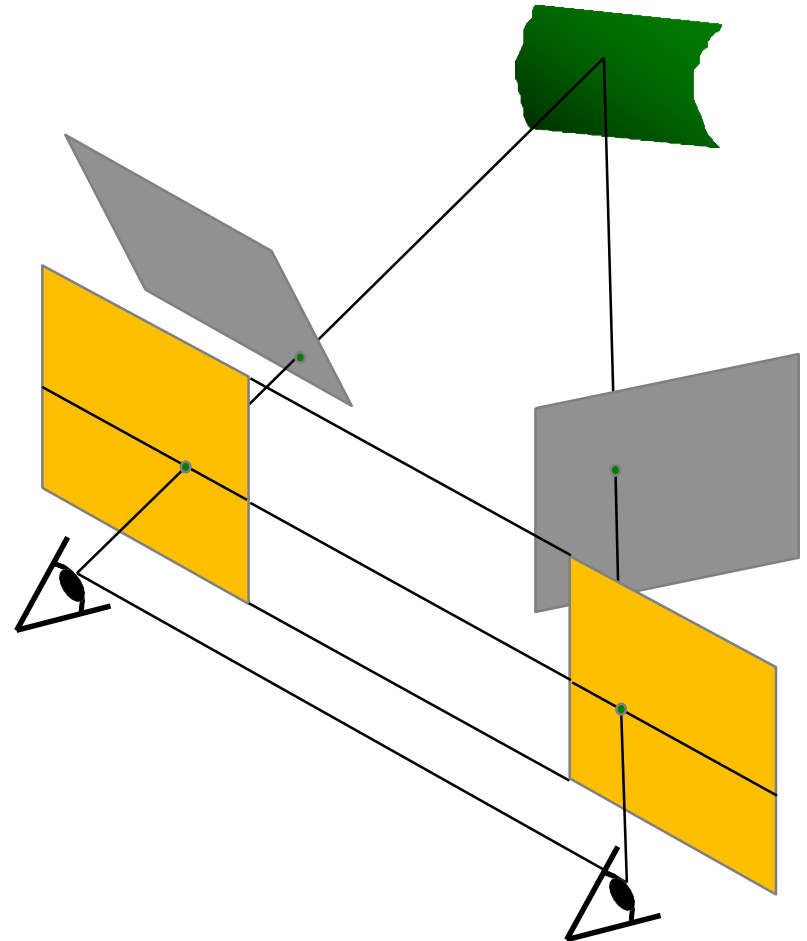


Image Rectification

- Re-project image planes onto a common plane parallel to the line between optical centers.
- Pixel motion is **horizontal** after this transformation.

(The testing images in this assignment have been rectified.)



Disparity Estimation

- After rectification, stereo matching becomes the **disparity estimation** problem.
- Disparity = horizontal displacement of corresponding points in the two images
 - Disparity of $\times = x_L - x_R$



- You need to implement Disparity Estimation in hw4.

Disparity Estimation

- “Hello world” algorithm: block matching
 - Consider SSD (Sum of Squared Distance) as matching cost

d	0	1	2	3	...	33	...	59	60
SSD	100	90	88	88	...	12	...	77	85

Minimal cost [Winner take all (WTA)]

$Image_L$

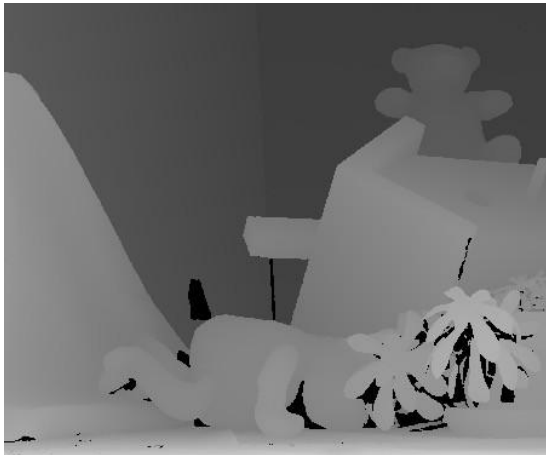


$Image_R$

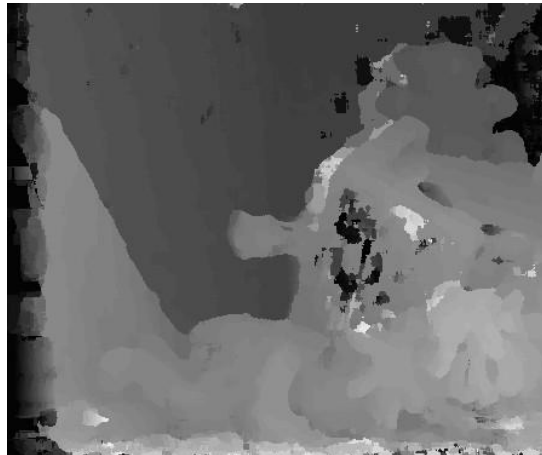


Disparity Estimation

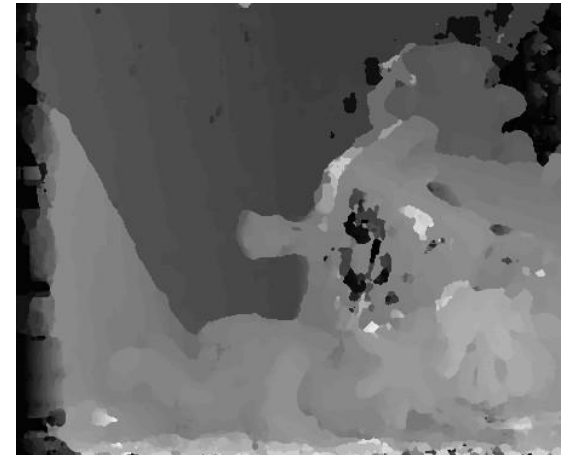
- Block matching result



Ground-truth



Window 5x5



After 3x3 median filter

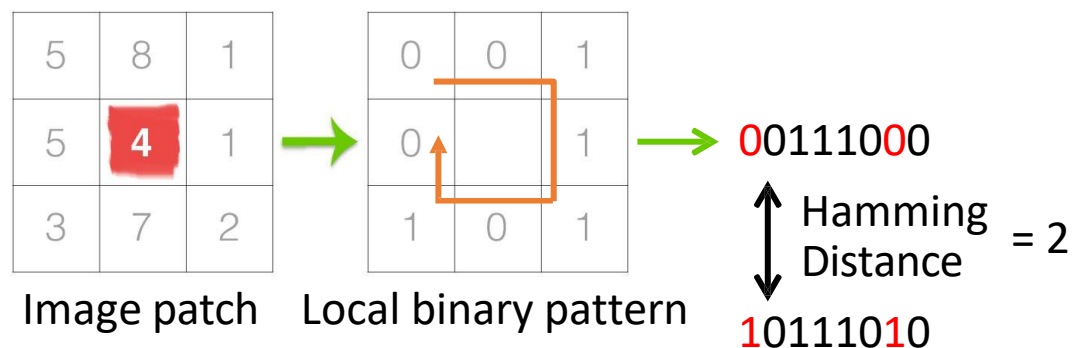
Typical Improved Pipeline

- It consists of 4 steps:
 - Cost computation
 - Cost aggregation
 - Disparity optimization
 - Disparity refinement

Step 1: Cost Computation

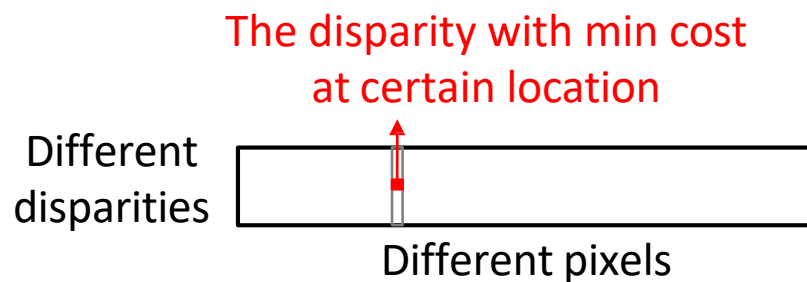
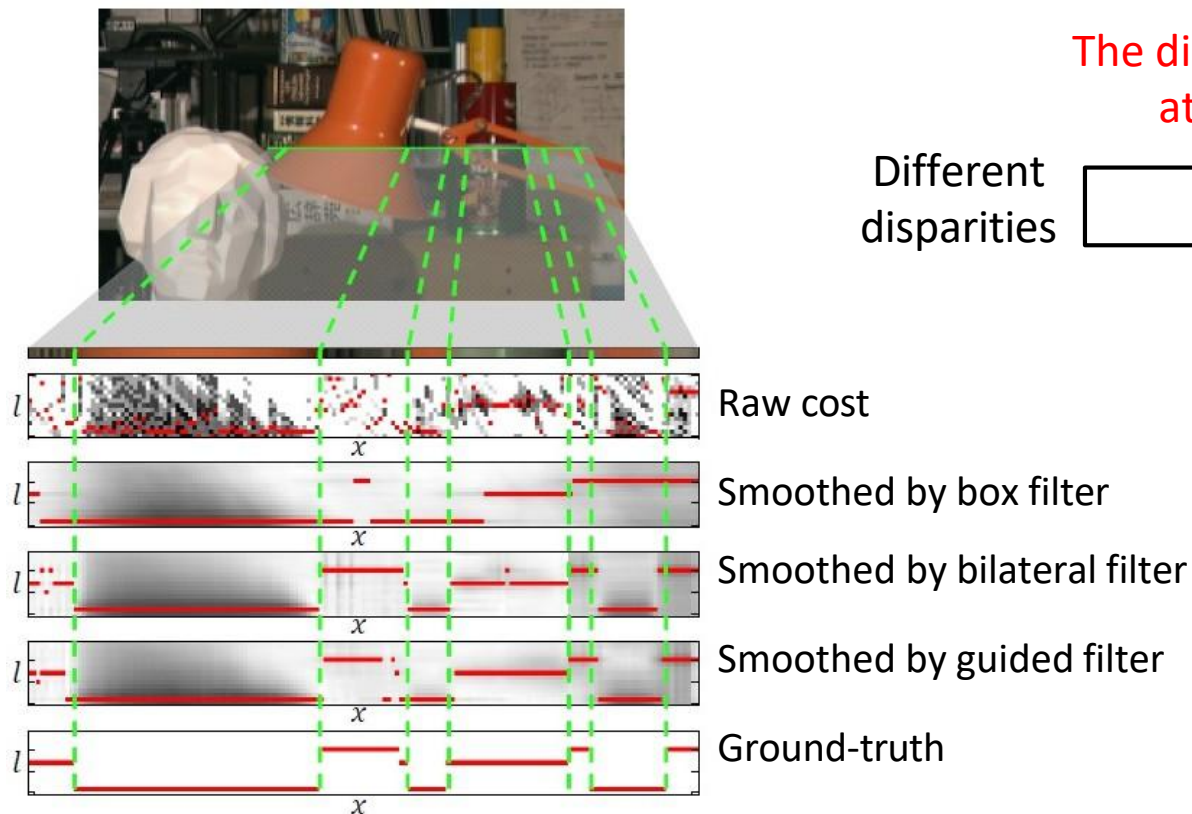
- Matching cost

- Squared difference (SD): $(I_p - I_q)^2$
- Absolute difference (AD): $|I_p - I_q|$
- Census cost



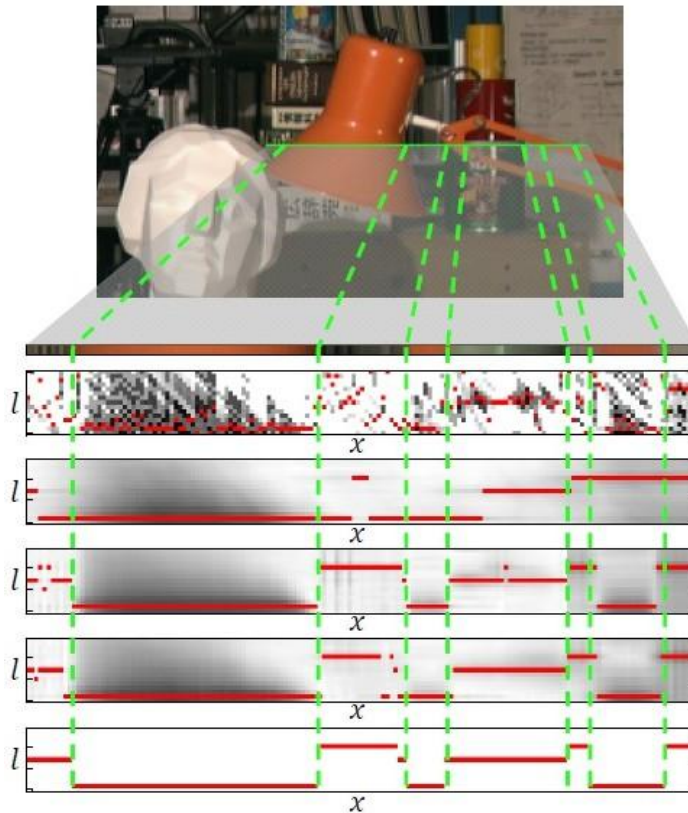
Step 2: Cost Aggregation

- Illustration of the matching cost



Step 3: Disparity optimization

- Winner-take-all



Step 4: Disparity Refinement

- Left-right consistency check
 - Compute disparity map D_L for left image
 - Compute disparity map D_R for right image
 - Check if $D_L(x, y) = D_R(x - D_L(x, y), y)$
 - If Yes, keep the computed disparity
 - If No, mark **hole** (invalid disparity)

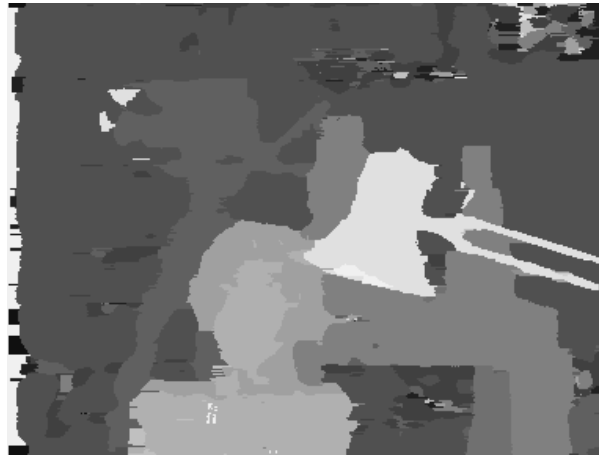
*Note: D_R are only used in this step!!
Only need to keep D_L for the next step.*



 
Two corresponding
positions in images

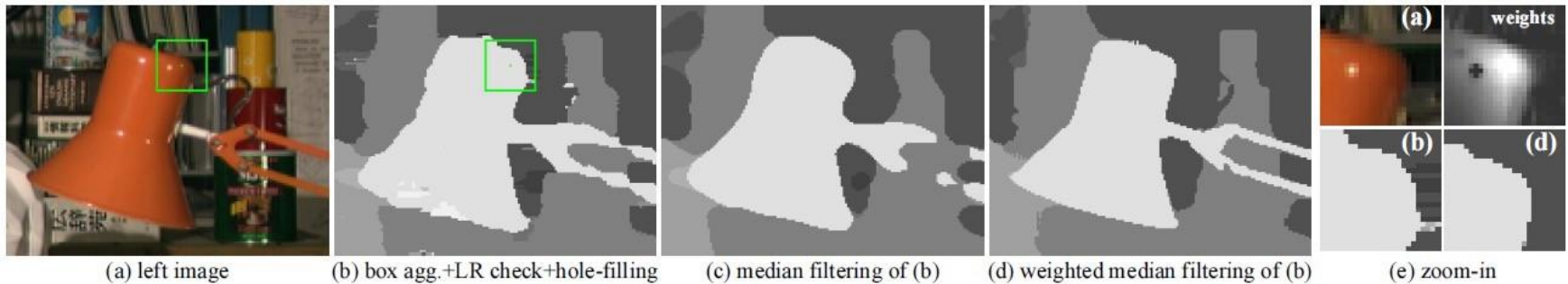
Step 4: Disparity Refinement

- Hole filling
 - F_L , the disparity map filled by closest valid disparity from left
 - F_R , the disparity map filled by closest valid disparity from right
 - Final filled disparity map $D = \min(F_L, F_R)$ (pixel-wise minimum)
 - Tips: pad maximum for the holes in boundary (start point)



Step 4: Disparity Refinement

- Weighted median filtering



Assignment Description(1/2)

- computeDisp.py (TODO)

```
import numpy as np
import cv2.ximgproc as xip

def computeDisp(I1, I2, max_disp):
    h, w, ch = I1.shape
    labels = np.zeros((h, w), dtype=np.float32)
    I1 = I1.astype(np.float32)
    I2 = I2.astype(np.float32)

    # >>> Cost Computation
    # TODO: Compute matching cost
    # [Tips] Census cost = Local binary pattern -> Hamming distance
    # [Tips] Set costs of out-of-bound pixels = cost of closest valid pixel
    # [Tips] Compute cost both I1 to I2 and I2 to I1 for later left-right consistency

    # >>> Cost Aggregation
    # TODO: Refine the cost according to nearby costs
    # [Tips] Joint bilateral filter (for the cost of each disparity)

    # >>> Disparity Optimization
    # TODO: Determine disparity based on estimated cost.
    # [Tips] Winner-take-all

    # >>> Disparity Refinement
    # TODO: Do whatever to enhance the disparity map
    # [Tips] Left-right consistency check -> Hole filling -> Weighted median filtering

    return labels.astype(np.uint8)
```

Good News:

you **CAN** use cv2.ximgproc package with plenty of filtering operations

Maximum possible disparity
(do not need to search the disparity larger than it)

You are not forced or limited to those tips. *But, they are good for you to improve your algorithm.*

CANNOT use deep learning based methods.

Assignment Description(2/2)

- main.py (completed)
 - Read image, execute stereo matching, and visualize disparity map.
 - Usage: `python3 main.py --image {input_image}`
- eval.py (**DO NOT EDIT this file**)
 - Compute disparity maps of the **left image** for the four standard test pairs from [Middlebury v2](#)

Tsukuba



with gt

Venus



without gt

Teddy



with gt

Cones



without gt

- Evaluation metric: bad pixel ratio (error threshold = 1)

Environment

- Python 3.6+
- Package
 - (optional)conda create -n <env name> python=3.6
 - pip3 install -r requirement.txt

Bonus

- Core Objective
 - Leverage Stereo Matching algorithms to implement depth perception in real-world scenarios and practical applications.
- Technical Challenges
 - Develop and implement a complete technical pipeline, ranging from Image Rectification to Disparity Map Computation.
- Deliverables
 - The final submission must include the algorithm source code, a technical report, and a demonstration video showcasing the dynamic results.

Submission(1/2)

- Directory architecture:
 - + R12345678_final/
 - computeDisp.py
- Put all the files in a directory named **StudentID_final** and compress the directory into zip file (named **StudentID_final.zip**)
- After TAs run “unzip R12345678_final.zip”, it should generate one directory named “R12345678_final”.
- Do **NOT** copy homeworks (including code and report) from others

Submission(1/2)

- Please submit those **two files**, i.e. your StudentID_final.zip and report.pptx, **separately** to NTU COOL
- Deadline: **6/5 11:59 pm**
- **Late submissions will not be accepted.**

文件上傳

上傳檔案，或者選擇已上傳的檔案。

 上傳文件

 使用網路攝影機

+ 新增另一個檔案

按一下此處，以找到已上傳的檔案

Grading (Total 105%)

- Performance: 80%

(Each of the four images must satisfy its respective bad pixel ratio threshold.)

Tsukuba	Venus	Teddy	Cones
< 8	< 5	< 18	< 15

- Ranking: $(\prod_{i=1}^4 \text{bad pixel ratio}_i) * \text{Total execution time}$
- A penalty of 30 points of the score will be deducted if the execution time for any image exceeds 10 minutes.
- A penalty of 20 points will be deducted if any of the four images fails to meet its respective bad pixel ratio threshold.
- Bouns: 25%

TA information

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